Q452: a single-box weight is a transfer matrix, and its output obeys a linear recurrence the target violates

A capacity refutation of every *configuration-determined* weight at $(k, \ell) = (1, 0)$, an explicit in-domain witness at $(k, \ell) = (2, 0)$, and the refutation of Conjecture Q434

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Abstract

Q430 refuted the *multiplicity-determined* single-box weights and named its own repair: the weight must read the cyclic index. Q452 asks whether such an *index-aware* weight exists. We answer by changing coordinates. A single box at cyclic slot i is legal exactly when $m_{i-1}(x) \geq 1$, and it reads precisely that multiplicity, decrementing it and incrementing m_i; so a cylindric tableau of content (1^n) **is** a length-n walk of w chips hopping clockwise on the cycle $\mathbb{Z}_h$, started with all w chips at site h (Lemma 5). In those coordinates the HKKO sign is *not* a function of the endpoint partition at all but of the final chip configuration: $c^- = +1$ when m is supported on even sites, -1 when on odd sites, 0 otherwise (Lemma 6).

Consequently every weight that reads only the local configuration — a class strictly containing the index-aware weights — is a **transfer matrix**, and $\sum_T c^-(T) W_T = \epsilon^{\mathsf{T}} A^n \delta_0$ (Theorem 8). Two things follow with no solving. First, A is $\mathbb{Z}_h$-graded, which *proves* the observed vanishing at odd n (Theorem 11). Second, the output sequence obeys a linear recurrence whose order is bounded by the smallest graded dimension (Theorem 12). At $(k, \ell) = (1, 0)$ that dimension is **1**, forcing the *identity* $\Sigma_2 \Sigma_6 = \Sigma_4^2$; the Warnaar targets give $c_2 c_6 - c_4^2 = (t-1)^4 (t+1)(t^2+t+1) \neq 0$. **No configuration-determined weight exists at** $(k, \ell) = (1, 0)$ (Theorem 13). The same machine kills a *second, structurally different* fibre family: at $(k, \ell) = (1, 1)$ the smallest graded dimension is 2, so the forced relation is the vanishing of the 3×3 Hankel determinant of $(\Sigma_2, \ldots, \Sigma_{10})$ — and the targets there give $-(t-1)^3 (t+1)^3 (t^2+1) \neq 0$ (Theorem 15). Two refutations, two different orders of recurrence, so the first is not an accident of the rank-one degeneracy.

The result is **unbundled** from two things it does not say. (i) $(k, \ell) = (1, 0)$ is *outside* Korff's domain $h > 2$; inside it the same capacity relation first appears at $n = 14$, and the Warnaar engine died at $n = 7$, so $k \geq 2$ **is undecided** and we print the n we reached. (ii) In-domain at $(k, \ell) = (2, 0)$ an index-aware weight with values in $\mathbb{Q}[t]$ *does* reproduce $n = 2, 4, 6$; it is exhibited (Proposition 17). Separately, the measured degrees refute Conjecture Q434 of the Q430 paper (§10), and a derived degree law at $k = 1$ matches 9/9 while failing 5/8 off its derivation — a firing negative control.

§0 Holdings and instruments, measured at the start of this slot

object	address	state
HKKO 2301.13117 LaTeX source	`library/sources/20261009pm-cit/warnaar-comment-HKKO-2301.13117.tex`	on disk, 23 317 B
Ishikawa–Wakayama MSF III source	copied today `/tmp/browse/src-0312358` → `library/sources/src-0312358/` (11 files)	**rescued from `/tmp`**
`/tmp/browse/iw-amsf.pdf`	—	**196 B**: an Apache 404 wearing `.pdf`. **Not copied.**
Warnaar engine (M1)	`proofs/code-1010-q410/{core,warn}.py`	on disk, importable
This session's code	`proofs/code-1014-q451/`, `proofs/code-1014-q452/`	written and run today
SageMath	—	**not installed**; pure Python + SymPy

Instrument failures this slot, recorded because they bound the results. The Warnaar ground-truth job **died** (no process, log truncated, almost certainly memory) during $k = 2, \ell = 0, n = 7$; the symbolic 4×4 Hankel job at $(h, w) = (4, 2)$ likewise printed its 2×2 and 3×3 lines and then died. Both are reported as reached-n bounds in §9 rather than as absences.

1 Definitions, fixed once

Throughout $k \geq 1$, $\ell \geq 0$, and $h := 2k$, $w := 2\ell + 2$. Korff's 2011 domain is $h > 2$ and $n \leq w$; membership is flagged at every fibre. We keep the Q430 conventions verbatim.

Definition 1 (region, configuration, cylindric tableau)**.** $R := \{x \in \mathbb{Z}^h : x_1 \geq \cdots \geq x_h \geq x_1 - w\}$; for $x \in R$, $m_i(x) := x_i - x_{i+1}$ $(1 \leq i \leq h-1)$ and $m_h(x) := w - (x_1 - x_h)$, so $m_i \geq 0$ and $\sum_{i \in \mathbb{Z}_h} m_i = w$. A *cylindric tableau of content* (1^n) is a sequence $T = (x^{(0)}, \ldots, x^{(n)})$ in R with $x^{(0)} = 0$ and $x^{(a)} - x^{(a-1)} = e_{i_a}$; we write $\mathcal{T}(k, \ell, (1^n))$, $\lambda(T) := x^{(n)}$, and $m^{(a)} := m(x^{(a)})$.

Definition 2 (HKKO sign)**.** For λ of length $\leq h$: $c^-_{h,w}(\lambda) := +1$ if $\lambda_{2i-1} = \lambda_{2i}$ for $1 \leq i \leq k$; -1 if $\lambda_1 - \lambda_h = w$ and $\lambda_{2i} = \lambda_{2i+1}$ for $1 \leq i \leq k-1$; 0 otherwise. Set $c^-(T) := c^-_{h,w}(\lambda(T))$.

Definition 3 (the three nested weight classes)**.** A *multiplicative weight* is $\omega : R \times \{0,1\}^h \to \mathbb{Q}[t]$ with $W_T := \prod_{a=1}^{n} \omega(x^{(a-1)}, \theta^{(a)})$. It is

(M) *multiplicity-determined* if $\omega(x, e_i) = f(m_{i-1}(x))$ for a single f;

(I) *index-aware* if $\omega(x, e_i) = g\big(i, m_{i-1}(x)\big)$ — **the class Q452 asks about**;

(C) *configuration-determined* if $\omega(x, e_i) = \Omega\big(m(x), i\big)$ for a single function Ω, i.e. the weight reads the whole local multiplicity vector but **nothing else**.

Plainly (M) $\subsetneq$ (I) $\subseteq$ (C). Note the full class is strictly larger than (C): since $\sum_i x_i^{(a)} = a$, the vector x carries, beyond $m(x)$, exactly the *step index* a. A weight outside (C) is one that reads how far along the tableau it is.

Specification 4 (the identity under test, unchanged from Q430)**.** $\displaystyle\sum_{T \in \mathcal{T}(k,\ell,(1^n))} c^-(T)\, W_T(t) = c_{(1^n)}(t)$, where $c_\alpha(t)$ is the coefficient of x^α in the Warnaar determinant $G \cdot (x_1 \cdots x_n)^h$ (engine M1).

2 The chip reformulation

Lemma 5 (legality, and what a box reads)**.** *Let $x \in R$ and $i \in \mathbb{Z}_h$. Then $x + e_i \in R$* ***if and only if*** *$m_{i-1}(x) \geq 1$, and in that case*

$$m_{i-1}(x+e_i) = m_{i-1}(x) - 1, \qquad m_i(x+e_i) = m_i(x) + 1, \qquad m_j(x+e_i) = m_j(x) \ (j \neq i-1, i).$$

Moreover Korff's cyclic index set is $J(e_i) = \{i - 1 \bmod h\}$, so the box reads exactly $m_{i-1}(x)$.

Proof. Write $y = x + e_i$. For $2 \leq i \leq h$: the only weak-decrease constraint that can fail is $y_{i-1} \geq y_i$, i.e. $x_{i-1} \geq x_i + 1$, i.e. $m_{i-1}(x) \geq 1$; the constraint $y_i \geq y_{i+1}$ holds since $x_i \geq x_{i+1}$, and $y_h \geq y_1 - w$ holds because $y_1 = x_1$ and $y_h \geq x_h$. For $i = 1$: weak decrease is automatic ($y_1 = x_1 + 1 \geq x_1 \geq x_2 = y_2$), and the cylindric constraint $y_h \geq y_1 - w$ reads $x_h \geq x_1 + 1 - w$, i.e. $w - (x_1 - x_h) \geq 1$, i.e. $m_h(x) = m_0(x) \geq 1$. The displayed updates are immediate from $m_j = x_j - x_{j+1}$ and $m_h = w - (x_1 - x_h)$, in both the case $i \leq h - 1$ and $i = h$ (where m_{h-1} drops and m_h rises) and $i = 1$ (where m_h drops and m_1 rises). Finally $J(\theta) = \{j : \theta_j = 0, \theta_{j+1} = 1\}$ gives $J(e_i) = \{i - 1\}$. □

So, writing $\Omega_{h,w} := \{m \in \mathbb{Z}^h_{\geq 0} : \sum_i m_i = w\}$, with $|\Omega_{h,w}| = \binom{w+h-1}{h-1}$:

a cylindric tableau of content (1^n) is a length-n walk of w indistinguishable chips hopping clockwise, one chip one site per step, on the cycle $\mathbb{Z}_h$, started at $m^{(0)} = (0, \ldots, 0, w)$; the step moving a chip off site j reads the number of chips then sitting at j.

3 The sign is a function of the configuration

Lemma 6 (the sign is configurational)**.** *For every $T \in \mathcal{T}(k, \ell, (1^n))$,*

$$c^-(T) = \epsilon\big(m^{(n)}\big), \qquad \epsilon(m) := \begin{cases} +1, & \operatorname{supp}(m) \subseteq \{2, 4, \ldots, h\} \quad (\textit{even sites}), \\ -1, & \operatorname{supp}(m) \subseteq \{1, 3, \ldots, h-1\} \quad (\textit{odd sites}), \\ 0, & \textit{otherwise,} \end{cases}$$

and the two cases are mutually exclusive.

Proof. Put $\lambda = \lambda(T)$ and $m = m^{(n)}$. For $1 \leq i \leq k$ we have $2i - 1 \leq h - 1$, so $m_{2i-1} = \lambda_{2i-1} - \lambda_{2i}$; hence the $+1$ condition "$\lambda_{2i-1} = \lambda_{2i}$ for all $1 \leq i \leq k$" says exactly that $m_1 = m_3 = \cdots = m_{h-1} = 0$, i.e. $\operatorname{supp}(m)$ misses every odd site. For the -1 condition: $\lambda_1 - \lambda_h = w$ is $m_h = 0$, and for $1 \leq i \leq k - 1$ we have $2i \leq h - 2$, so $m_{2i} = \lambda_{2i} - \lambda_{2i+1}$ and the condition "$\lambda_{2i} = \lambda_{2i+1}$" says $m_2 = m_4 = \cdots = m_{h-2} = 0$; together with $m_h = 0$ this is exactly that $\operatorname{supp}(m)$ misses every even site. Exclusivity: both would force $m = 0$, contradicting $\sum_i m_i = w \geq 2$. □

Remark 7. This is the step that makes the whole paper possible, and it is a *coordinate* fact, not a new theorem: the sign looked like a global property of the endpoint partition and is in fact a local property of the final chip configuration. Verified independently against the Q430 implementation of $c^-_{h,w}(\lambda)$ on all reachable states for $(h, w) \in \{(2,2), (4,2), (6,2), (2,4), (4,4), (6,4), (4,6)\}$ and $1 \leq n \leq 6$: 125 **states,** 0 **mismatches**.

4 Theorem A: a local weight is a transfer matrix

Theorem 8 (transfer matrix)**.** *Let ω be configuration-determined (class (C) of Definition 3); in particular let it be index-aware. Define $A \in \mathbb{Q}[t]^{\Omega_{h,w}\times\Omega_{h,w}}$ by*

$$A_{m',m} := \begin{cases} \Omega(m,i) & \text{if } m' = m - e_{i-1} + e_i \text{ for some } i \in \mathbb{Z}_h \text{ with } m_{i-1} \geq 1, \\ 0 & \text{otherwise,} \end{cases}$$

let δ_0 be the indicator of $m^{(0)} = (0,\dots,0,w)$ and ϵ the row vector of Lemma 6. Then for all $n \geq 0$

$$\Sigma_n \;:=\; \sum_{T\in\mathcal{T}(k,\ell,(1^n))} c^-(T)\, W_T(t) \;=\; \epsilon^{\mathsf{T}} A^n \delta_0.$$

Proof. By Lemma 5 the map $T \mapsto (m^{(0)},\dots,m^{(n)})$ is a bijection from $\mathcal{T}(k,\ell,(1^n))$ onto the set of length-n walks in the chip graph started at $m^{(0)}$: the walk determines the box slots i_a (the unique i with $m^{(a)} = m^{(a-1)} - e_{i-1} + e_i$), hence $x^{(a)} = \sum_{b\leq a} e_{i_b}$, and conversely. Under this bijection $W_T = \prod_a \Omega(m^{(a-1)}, i_a) = \prod_a A_{m^{(a)},m^{(a-1)}}$, because a configuration-determined weight reads only $m^{(a-1)}$ and i_a. Summing the products of entries over all walks of length n from $m^{(0)}$ to m gives $(A^n\delta_0)_m$, and weighting by $c^-(T) = \epsilon(m^{(n)})$ (Lemma 6) and summing over m gives the claim. □

Remark 9 (what this does *not* cover). The hypothesis is exactly that the weight is *memoryless*. A weight that also reads the step index $a = \sum_i x_i^{(a-1)}$ gives a time-dependent product $A_n \cdots A_1$ and escapes every consequence below. That escape is the specification this paper hands forward (§11).

Remark 10 (verification). $\epsilon^{\mathsf{T}} A^n \delta_0$ was checked against direct enumeration of $\mathcal{T}(k,\ell,(1^n))$ with symbolic $g_{i,m}$, for $(h,w) \in \{(2,2),(4,2),(4,4)\}$ and $n = 1,\dots,6$: 18 **of** 18 **identical as polynomials**. The direct enumerator was separately checked against the published Q430 table at $(k,\ell,n) = (2,1,4)$: $|\mathcal{T}| = 10$, four tableaux with $c^- \neq 0$, $\sum_T c^-(T) = 2$, and the four surviving monomials $g_{4,4}g_{1,1}g_{2,1}g_{3,1}$, $g_{4,4}g_{1,1}^2 g_{4,3}$, $g_{4,4}g_{4,3}g_{1,2}g_{1,1}$, $-g_{4,4}g_{4,3}g_{4,2}g_{4,1}$ — the registry equation, reproduced term by term.

5 Theorem B: the grading, and why the odd fibres vanish

Theorem 11 ($\mathbb{Z}_h$-grading)**.** *Let $\Phi(m) := \sum_{i=1}^h i\, m_i$. Then A maps the span V_p of $\{m : \Phi(m) \equiv p\}$ into V_{p+1}, and $\Phi(m^{(n)}) \equiv n \pmod h$. Consequently $\epsilon^{\mathsf{T}} A^n \delta_0 = 0$ for every odd n — i.e. $\Sigma_n = 0$* ***identically for odd n, for every configuration-determined weight****.*

Proof. A move $j \to j+1$ with $j < h$ changes Φ by $+1$; the move $h \to 1$ changes it by $1 - h \equiv 1 \pmod h$. So each step raises Φ by 1 mod h, and $\Phi(m^{(0)}) = h \cdot w \equiv 0$, giving $\Phi(m^{(n)}) \equiv n$. Now suppose $\epsilon(m) \neq 0$. If $\mathrm{supp}(m)$ lies in the even sites then every term $i\, m_i$ is even, so $\Phi(m) \equiv 0 \pmod 2$. If $\mathrm{supp}(m)$ lies in the odd sites then $\Phi(m) \equiv \sum_{i \text{ odd}} m_i = w \equiv 0 \pmod 2$, since $w = 2\ell+2$. As h is even, $n \equiv \Phi(m^{(n)}) \pmod 2$; hence ϵ vanishes on every state reachable in an odd number of steps. □

This proves, rather than observes, the zeros in the Q430 family table: all nine vacuous fibres there have n odd. It is a *consistency* check on the architecture, not an obstruction — the targets $c_{(1^n)}$ vanish at odd n too, on all 14 measured rows.

6 Theorem C: the capacity bound

Theorem 12 (capacity)**.** *Let* $\rho := \min_{p\in\mathbb{Z}_h} \dim V_p$*. For each residue* r *the subsequence* $\big(\Sigma_{r+jh}\big)_{j\ge 1}$ *satisfies a linear recurrence over* $\mathbb{Q}(t)$ *of order at most* ρ*, with a characteristic polynomial* independent of r*. In particular the whole sequence* $(\Sigma_n)_n$ *satisfies a linear recurrence of order* $\le |\Omega_{h,w}|$ *whose characteristic polynomial is a polynomial in* z^h*.*

Proof. $B := A^h|_{V_0}$ factors as $V_0 \to V_1 \to \cdots \to V_{h-1} \to V_0$, hence $\operatorname{rank} B \le \min_p \dim V_p = \rho$ and the characteristic polynomial of B is $z^{\dim V_0 - \rho'} q(z)$ with $\rho' = \operatorname{rank} B \le \rho$ and $\deg q = \rho'$. Writing $\Sigma_{r+jh} = \big(\epsilon^{\mathsf{T}} A^r\big) B^j \delta_0$ for $\delta_0 \in V_0$, the Cayley–Hamilton relation for B applied to $B^j\delta_0$ with $j \ge 1$ gives the order-ρ' recurrence with characteristic polynomial q, the same q for every r. □

This is a *capacity* statement in the Q405 sense: it bounds what the architecture can emit before any weight is chosen.

7 Theorem D: the refutation at $(k, \ell) = (1, 0)$

Theorem 13 (no configuration-determined weight at $(k,\ell) = (1,0)$)**.** *Let* $(k,\ell) = (1,0)$*, so* $h = 2$ *and* $w = 2$*. For* ***every*** *configuration-determined weight,*

$$\Sigma_{2r} = \Sigma_2 \cdot v^{\,r-1} \quad (r \ge 1) \quad \textit{for a single } v \in \mathbb{Q}(t), \qquad \textit{hence} \qquad \Sigma_2\,\Sigma_6 = \Sigma_4^2 \ \textit{identically.} \tag{1}$$

But the Warnaar targets are $c_2 = 1 - t$*,* $c_4 = (t-1)^2(t+2)$*,* $c_6 = -(t-1)^3(t^3+3t^2+6t+5)$*, and*

$$\boxed{\; c_2c_6 - c_4^2 \;=\; (t-1)^4(t+1)(t^2+t+1) \;\neq\; 0\ . \;}$$

Therefore ***no configuration-determined weight — in particular no index-aware weight, and a fortiori no multiplicity-determined weight — satisfies Specification 4 on the fibres*** $\alpha = (1^n)$ ***at*** $(k,\ell) = (1,0)$***.***

Proof. Here $\Omega_{2,2} = \{(2,0),(1,1),(0,2)\}$ and $\Phi(2,0) = 2$, $\Phi(1,1) = 3$, $\Phi(0,2) = 4$, so modulo $h = 2$ we get $\dim V_0 = 2$ (states $(2,0),(0,2)$) and $\dim V_1 = 1$ (state $(1,1)$). Thus $\rho = 1$ and, in the notation of Theorem 12, $B = A^2|_{V_0}$ is a 2×2 matrix factoring through a one-dimensional space, so $\operatorname{rank} B \le 1$ and $\det B = 0$. By Cayley–Hamilton $B^2 = (\operatorname{tr} B)B$, hence $B^r = v^{r-1}B$ with $v := \operatorname{tr} B$ for all $r \ge 1$. Since $\delta_0 = \delta_{(0,2)} \in V_0$ and $A^{2r}\delta_0 \in V_0$, $\Sigma_{2r} = \epsilon^{\mathsf{T}} B^r \delta_0 = v^{r-1}\epsilon^{\mathsf{T}} B\delta_0 = v^{r-1}\Sigma_2$, which is (1); then $\Sigma_2\Sigma_6 = \Sigma_2 \cdot v^2\Sigma_2 = (v\Sigma_2)^2 = \Sigma_4^2$. If Specification 4 held at $n = 2, 4, 6$ we would get $c_2c_6 = c_4^2$, and the boxed computation contradicts it (the right side is a nonzero polynomial; e.g. it takes the values 1 at $t = 0$ and 21 at $t = 2$). □

Remark 14 (the identity (1) is not an artefact of $h = 2$ alone). $\Sigma_2\Sigma_6 - \Sigma_4^2$ was expanded symbolically in the free variables $g_{i,m}$ for five pairs (h,w). It vanishes identically **exactly where** $\rho = 1$:

(h,w)	(k,ℓ)	$\lvert\Omega\rvert$	ρ	$\Sigma_2\Sigma_6 - \Sigma_4^2 \equiv 0$?
$(2,2)$	$(1,0)$	3	**1**	**yes**
$(4,2)$	$(2,0)$	10	2	no (30 terms)
$(6,2)$	$(3,0)$	21	3	no (31 terms)
$(2,4)$	$(1,1)$	5	2	no (6 terms)
$(4,4)$	$(2,1)$	35	8	no (20 terms)

A relation that held everywhere would be evidence of a bug in the expander; one that holds exactly on the row the rank argument singles out is evidence of the argument.

8 Theorem D′: a second refutation, at $(k,\ell)=(1,1)$, at recurrence order two

Theorem 15. *Let $(k,\ell)=(1,1)$, so $h=2$ and $w=4$. Here $|\Omega_{2,4}|=5$ with graded dimensions $(\dim V_0,\dim V_1)=(3,2)$, so $\rho=2$ and Theorem 12 forces*

$$\det\begin{pmatrix}\Sigma_2 & \Sigma_4 & \Sigma_6\\ \Sigma_4 & \Sigma_6 & \Sigma_8\\ \Sigma_6 & \Sigma_8 & \Sigma_{10}\end{pmatrix}=0 \qquad \textit{identically, for every configuration-determined weight.}$$

The Warnaar targets are $c_2=1$, $c_4=2-t^2$, $c_6=t^4-5t^2+5$, $c_8=7t^4-20t^2+14$ and $c_{10}=-(t^2-2)(t^8+2t^6+4t^4-27t^2+21)$, whose 3×3 Hankel determinant is

$$\boxed{-(t-1)^3(t+1)^3(t^2+1)\ \neq\ 0}$$

(value 1 at $t=0$, -5120 at $t=3$). Hence ***no configuration-determined weight satisfies Specification 4 on the fibres $\alpha=(1^n)$ at $(k,\ell)=(1,1)$ either.***

Proof. The states are (m_1,m_2) with $m_1+m_2=4$ and $\Phi(m)=m_1+2m_2\equiv m_1 \pmod 2$, giving $V_0=\{(4,0),(2,2),(0,4)\}$ and $V_1=\{(3,1),(1,3)\}$; so $\rho=2$ and, by Theorem 12, $(\Sigma_{2r})_{r\geq 1}$ obeys a linear recurrence of order ≤ 2 from $r=1$ onwards, which is exactly the vanishing of the displayed 3×3 Hankel. (Checked independently: the Hankel was expanded symbolically in the free variables $g_{i,m}$ and is the zero polynomial, while the 2×2 Hankel on $\Sigma_2,\Sigma_4,\Sigma_6$ is not — so the order is exactly 2, not less.) The boxed value is a direct computation. □

Remark 16 (why this matters more than a second data point). Theorem 13 rests on $\rho=1$, where the forced relation is the degenerate one ("the sequence is geometric"). One could reasonably suspect that the whole effect is an artefact of that degeneracy. Theorem 15 runs at $\rho=2$: a different fibre family, a different w, a different order of recurrence, a different Hankel size, a different discriminant — and the same verdict. The **firing** two-mechanism control is strongest here too: at $(k,\ell)=(1,1)$ the targets have $c_{(1^n)}(1)=1\neq 0$ for $n=2,4,6,8,10$, and the chip enumerator independently returns $\sum_T c^-(T)=1$ on fibres of size $2,6,18,54,162$ — **5 of 5, all non-vacuous**.

8.1 The satisfiability controls — and they localise the kill to $n=6$

A class-kill without a satisfiability control is indistinguishable from a broken fibre generator. Three were run.

Control 1 (Q430 reproduction). The index-aware weight $g(h,2)=1$, $g(1,1)=1$, $g(h,1)=t$ gives $\Sigma_2=1-t=c_{(1,1)}$ **exactly**, on a 2-tableau fibre, for $k=1,2,3,4$ — 4/4. The generator reproduces the published control.

Control 2 (the kill is the third equation, not the fibre). At $(k,\ell)=(1,0)$ the *polynomial* index-aware weight

$$g_{1,1}=2-t,\qquad g_{1,2}=-t^2,\qquad g_{2,1}=1,\qquad g_{2,2}=1 \qquad (\text{all in } \mathbb{Q}[t])$$

satisfies Specification 4 at $n=2$ and at $n=4$ **exactly** ($\Sigma_2=1-t$, $\Sigma_4=(t-1)^2(t+2)$), and fails at $n=6$ with residual

$$c_6-\Sigma_6=-(t-1)^3(t+1)(t^2+t+1).$$

So the fibres at $n = 2, 4$ are satisfiable in the narrow class, and Theorem 13 is testing the architecture at the first n where the recurrence bites. (That the residual carries the same factor $(t+1)(t^2+t+1)$ as the boxed discriminant is not a coincidence: by (1) the residual is $c_6 - c_4^2/c_2 = (c_2c_6 - c_4^2)/c_2$.)

Control 3 (in-domain satisfiability). See Proposition 17.

The targets were checked by a second mechanism. Every $c_{(1^n)}$ used here comes from the Warnaar determinant (engine M1). Its $t = 1$ specialisation must equal $\sum_T c^-(T)$ computed by the chip enumerator, which shares no code with M1: 14 **of** 14 **rows agree**. At $\ell = 0$ that control is *vacuous* (both sides are 0), and we say so; the **firing** instances are at $\ell = 1$, where $c_{(1^n)}(1) \in \{1, 1, 1, 1, 2, 4\}$ and the enumerator reproduces each. In particular M1 independently returns $c_{(1^4)} = 3 - t^2$ at $(k, \ell) = (2, 1)$, the published Q430 headline value.

9 The in-domain situation, stated with its boundary

Proposition 17 (an index-aware weight exists in-domain at $(k, \ell) = (2, 0)$ for $n \le 6$)**.** *At $(k, \ell) = (2, 0)$ — inside Korff's domain, $h = 4 > 2$ — the index-aware weight with values in $\mathbb{Q}[t]$*

$$g_{1,1} = -2(t-1), \quad g_{1,2} = 2t - 1, \quad g_{2,1} = t - 1, \quad g_{2,2} = \tfrac{1}{4}(9t^2 - 25t + 31),$$

$$g_{3,1} = 1 - t, \quad g_{3,2} = 0, \quad g_{4,1} = 1 - t, \quad g_{4,2} = 1$$

satisfies Specification 4 at $n = 2, 4, 6$ ***exactly****, against the targets $c_2 = 1 - t$, $c_4 = 3(t-1)^2$, $c_6 = (t-1)^3(t^2 - 2t - 14)$.*

Proof. Direct substitution into Σ_n for $n = 2, 4, 6$; all three residuals are the zero polynomial. (The witness was found structurally rather than by search: Σ_2 and Σ_4 do not involve $g_{2,2}$ at all, while Σ_6 is *linear* in it, so one solves Σ_2 and Σ_4 first and then Σ_6 linearly. Independently, a Gröbner basis for $\{\Sigma_n - c_n\}_{n=2,4,6}$ in the eight unknowns over $\mathbb{Q}(t)$ is not $\{1\}$, so the system is consistent — a second instrument agreeing with the exhibited witness.) □

This is outcome (2A) and we claim only it: a solution on the measured fibres $n \le 6$ at one in-domain (k, ℓ). It is not Warnaar's statistic, it is not a claim at $n \ge 8$, and it is not a claim at $\ell \ge 1$.

Why $k \ge 2$ is undecided, with the n printed. At $(h, w) = (4, 2)$ the graded dimensions are $(\dim V_0, \ldots, \dim V_3) = (3, 2, 3, 2)$, so $\rho = 2$ and the capacity relation of Theorem 12 is a 3×3 Hankel vanishing on $(\Sigma_2, \Sigma_6, \Sigma_{10}, \Sigma_{14}, \Sigma_{18})$ within one residue class; for the interleaved even sequence the first vanishing Hankel is 4×4 on $\Sigma_2, \ldots, \Sigma_{14}$. Measured symbolically: the 2×2 Hankel on $\Sigma_2, \Sigma_4, \Sigma_6$ does **not** vanish, nor does the 3×3 on $\Sigma_2, \ldots, \Sigma_{10}$ — consistent with order 4, and the 4×4 job died before printing. On the target side the Warnaar engine reached $n = 6$ at $k = 2$ (168 s) and **died at** $n = 7$. So:

Undecided for $k \ge 2$. Capacity relation needs $n = 14$; targets reached $n = 6$.

This is pre-registered outcome **(2D)**, and the domain is named rather than hinted at.

10 Conjecture Q434 is false

Proposition 18. *Conjecture Q434 of the Q430 paper — "*$\deg_t c_{(1^n)}(t) < w$ *on every fibre with* $c_{(1^n)} \neq 0$*" — is* ***false****, both outside and inside Korff's domain.*

Proof. Counterexamples from engine M1. At $(k,\ell) = (1,0)$, $w = 2$: $\deg_t c_{(1^4)} = 3$, $\deg_t c_{(1^6)} = 6$, $\deg_t c_{(1^8)} = 10$. In-domain at $(k,\ell) = (2,0)$, $w = 2$: $\deg_t c_{(1^4)} = 2$ and $\deg_t c_{(1^6)} = 5$. Each exceeds or equals $w = 2$. □

The conjecture was formed on the Q430 family table, which stops at $n \leq 5$; the degrees there are genuinely small, and the growth only shows up beyond the tabulated range. Nothing in Q430 collapses: the 15/15 divisibility failures stand on their own measurements. What dies is the *upgrade* from "15 of 15 tested" to "every non-vacuous fibre". **A conjecture read off the end of a table inherits the table's truncation.**

10.1 A derived degree law at $k = 1$, and a firing negative control

For $k = 1$ the Warnaar determinant is 1×1: $G = \sum_{y\in\mathbb{Z}} t^{My^2-y}\big(\dot{e}_{n-1+1-Ny} - \dot{e}_{n+1+1-Ny}\big)$ with $M = \ell + 2$, $N = 2M$. A term is nonzero only if some $\dot{e}_r$ with $0 \leq r \leq 2n$ occurs, which confines y to $-n/N \leq y \leq (n+2)/N$; and $y \mapsto My^2 - y$ is injective on $\mathbb{Z}$ for $M \geq 2$, so *distinct winding sectors cannot cancel against each other.* This predicts

$$\deg_t c_{(1^n)}(t) \;=\; \max\Big\{\, My^2 - y \;:\; y \in \mathbb{Z},\; -\tfrac{n}{N} \leq y \leq \tfrac{n+2}{N} \,\Big\},$$

provided the extremal sector contributes a nonzero coefficient of $x^{(1^n)}$.

k	ℓ	w	n	$\deg_t c_{(1^n)}$	predicted	
1	0	2	2	1	1	✓
1	0	2	4	3	3	✓
1	0	2	6	6	6	✓
1	0	2	8	10	10	✓
1	1	4	2	0	0	✓
1	1	4	4	2	2	✓
1	1	4	6	4	4	✓
1	1	4	8	4	4	✓
1	1	4	10	10	10	✓
negative control — off the derivation's domain ($k \geq 2$: the determinant mixes sectors)						
2	0	2	2	1	0	✗
2	0	2	6	5	4	✗
2	1	4	4	2	0	✗
3	0	2	2	1	0	✗
3	0	2	4	2	0	✗

9 of 9 on the derivation's domain; 3 of 8 off it. A formula that matched everywhere would be suspicious; one that matches exactly where its hypothesis holds is a control that fires.

Remark 19 (what this would give, and the gap that stops it). The predicted degree grows like $n^2/(4M)$ — **quadratically in** n — whereas any multiplicative weight drawn from a *finite* set of polynomials of degree $\leq d$ satisfies $\deg_t \Sigma_n \leq nd$, **linearly in** n. Were the law proved, it would refute at $k = 1$ for *every* ℓ and for the still wider class of time-dependent weights of bounded degree, by a pure capacity count. This is Q436 one rung further: Q436 showed Warnaar's t is quadratic

in the winding number; this says the winding number itself grows along the fibre family, so the quadratic outruns any product over boxes without needing to identify a slot at all. **The gap is exactly one statement**: that the extremal admissible y contributes a nonzero coefficient of $x^{(1^n)}$. Verified 9/9; **not proved**. Allocated as Q458. (The $n = 10$ row is the sharpest: the predicted degree jumps $4 \to 10$ between $n = 8$ and $n = 10$, and it does.)

11 Specification 6: what the theorem asks for next

Specification 20 (superseding Specification 5)**.** A multiplicative weight satisfying Specification 4 on the fibres $\alpha = (1^n)$ at $(k, \ell) = (1, 0)$ must **not** be configuration-determined: it must read, beyond the local multiplicity vector $m(x)$ and the slot i, the **step index** $a = \sum_i x_i$ — the number of boxes already placed. Equivalently, it must be a *time-dependent* transfer matrix A_a, not a fixed A.

Clause (i) of Specification 5 (read the cyclic index) is *necessary but far from sufficient*: index-awareness is a strict sub-class of (C), and all of (C) dies at $(1, 0)$. Clause (ii) (first-step divisibility, Q430 Theorem B) survives untouched — Lemma 5 re-proves the forced first step, since $m^{(0)}$ is supported on the single site h. So: **neither of the two conditions Q452 was sent to test is what does the killing.** The killer is a third thing, and it is a property of the *architecture* rather than of the weight: memorylessness forces a linear recurrence, and the target is not linearly recurrent at the order memorylessness allows.

12 Gaps, stated precisely

1. $k \geq 2$ **is undecided.** Theorems 13 and 15 are proved only at $(k, \ell) = (1, 0)$ and $(1, 1)$, both of which have $h = 2$ and so lies **outside** Korff's 2011 domain $h > 2$. (It is a legitimate instance of Specification 4, which is Warnaar's and holds for all $k \geq 1$; it is not an instance of Korff's construction.) The obstruction is computational and quantified in §9: relation at $n = 14$, targets at $n = 6$. **Q457**.

2. **The** $k = 1$ **degree law is measured, not proved**: non-vacuity of the extremal winding sector. **Q458**.

3. **Nothing here constrains time-dependent weights** (Specification 20), nor weights on fibres with content other than (1^n).

4. The symbolic 4×4 Hankel at $(h, w) = (4, 2)$ **did not complete**; the order-4 reading of the even sequence there is inferred from Theorem 12 plus the two non-vanishing Hankels actually computed, not measured directly.

Appendix: the targets used, all from engine M1

k	ℓ	n	$c_{(1^n)}(t)$	$c(1)$	$\sum_T c^-(T)$
1	0	2	$1-t$	0	0
1	0	4	$(t-1)^2(t+2)$	0	0
1	0	6	$-(t-1)^3(t^3+3t^2+6t+5)$	0	0
1	0	8	$(t-1)^4(t^6+4t^5+10t^4+20t^3+28t^2+28t+14)$	0	0
2	0	2	$1-t$	0	0
2	0	4	$3(t-1)^2$	0	0
2	0	6	$(t-1)^3(t^2-2t-14)$	0	0
3	0	2	$1-t$	0	0
3	0	4	$3(t-1)^2$	0	0
1	1	2	1	1	1
1	1	4	$2-t^2$	1	1
1	1	6	t^4-5t^2+5	1	1
1	1	8	$7t^4-20t^2+14$	1	1
1	1	10	$-(t^2-2)(t^8+2t^6+4t^4-27t^2+21)$	1	1
2	1	2	1	1	1
2	1	4	$3-t^2$	2	2
2	1	6	$5t^3-15t^2+14$	4	4

Last two columns: the two-mechanism control. All odd-n rows are 0 and are omitted; Theorem 11 explains them.